HYPER-LOCAL AVIATION WEATHER FORECAST VERIFICATION SYSTEM FOR SRI LANKAN MONSOON PATTERNS

COMMON INTEGRATED RESEARCH THESIS

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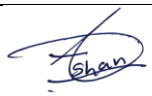
Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

A reliable weather system is crucial for safe operation at airports; however, it is often not stable and can rapidly vary during periods of monsoons and inter-monsoons in Sri Lanka. In this research project, we build an integrated and AI-based aviation weather forecasting and verification platform focusing on Bandaranaike International Airport (VCBI). The project comprises four integrated components: wind-speed and wind-direction forecasting along with their runway-specific interpretation and a pilot portal; QNH, dewpoint, and relative humidity computation; temperature and pressure forecasting combined with density altitude computation; and cloud and visibility forecasting. As described in the project documents, this project has historical data on VCBI for 2019-2026. There are 121,407 observations in our project database, and 51,175 observations remained after using the reported unified three-hour preprocessing pipeline. Each component employs different approaches depending on the target variable, such as XGBoost, Temporal Fusion Transformer, LightGBM, and the reported Random Forest–LSTM combination.

Most robust numerical results presented include: wind XGBoost MAE of 1.0516 knots, RMSE of 1.4404 knots and R^2 of 0.7798 using 10,235 test samples; dewpoint MAE of approximately 0.58 °C and R^2 of approximately 0.66; QNH MAE of approximately 1.8 hPa; and reported values of accuracy of 76.25%, 76.31% and 78.15% of cloud/visibility model for Decision Tree, Random Forest and XGBoost, respectively. Also, there is a reported hybrid temperature-pressure result of approximately R^2 0.84, but metric definitions and units need to be verified. They are not directly comparable as the target definitions, partitioning and evaluation procedures differ. The work presents a well-designed solution for the integration of airport-based predictions with deterministic aviation-related calculations, but it reveals some issues related to reproducibility, evidence of common interface, proper time-synchronized validation and validation by relevant stakeholders. It is an experimental research prototype and decision support system, not an aviation forecast certified for use nor a substitute for official weather products and professional opinion.

Keywords: aviation meteorology, hyper-local forecasting, machine learning, forecast verification, VCBI

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
DA	Density Altitude
ECMWF	European Centre for Medium-Range Weather Forecasts
ICAO	International Civil Aviation Organization
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
METAR	Meteorological Aerodrome Report
ML	Machine Learning
NWP	Numerical Weather Prediction
PA	Pressure Altitude
QNH	Altimeter setting referenced to mean sea level under aviation conventions
RH	Relative Humidity
RF	Random Forest
RMSE	Root Mean Squared Error
R^2	Coefficient of determination
TAF	Terminal Aerodrome Forecast
TFT	Temporal Fusion Transformer
VCBI	Bandaranaike International Airport ICAO location indicator
XGBoost	Extreme Gradient Boosting

1. INTRODUCTION AND PROJECT CONTEXT

1.1. Background and context

Weather is part of operational inputs in aviation and not merely a backdrop to operations. The wind impacts the performance of the airplane and the choice of runways; pressure and temperature impact the atmosphere and aircraft performance; humidity and dewpoint describe the state of moisture in the atmosphere; while clouds and visibility impact on the visual conditions in the environment. Weather information for aviation includes METAR reports and TAF forecasts.

Such a big prediction will be useful for regional planning; however, there could be some variations that won't be covered by the forecast. The topic assessment form highlights this problem as one applicable to tropical regions, where monsoon circulation, convection, and inter-monsoon transition might lead to sudden changes in the local weather situation. Motivation behind the project is to see if historical airport observations along with machine learning techniques can fill in the gaps in the current forecasts and verify them using observations.

The investigation centers on VCBI, Bandaranaike International Airport, Sri Lanka. However, the study is not limited to only one monsoon season. A typical scope of the project includes the Southwest Monsoon, Northeast Monsoon, and two inter-monsoon periods as the general season classification presented in the topic assessment form. Nevertheless, the seasonal interpretation has to be backed up by results labeled by particular season.

1.2 Stakeholders and user needs

The stakeholder analysis form mentions pilots, air traffic controllers, meteorological officers in the field of aviation, and operational managers at airports as stakeholders. The information requirements of each group of people are similar yet different. A user interface designed for pilots requires clear and quick-to-understand information, with the timestamp attached to it. A meteorological reviewer requires traceability of the source of observations, forecast time, time-stamped input data, model, and verification details.

Table 1.1.Stakeholder needs and design implications

Stakeholder	Potential need	Design implication
Pilots	Time-specific wind and weather context; runway-relative interpretation	Display forecast target time, units, source status and advisory limitations
Air traffic controllers	Consistent and interpretable local weather information	Use clear timestamps and avoid ambiguous mixed-source values
Aviation meteorological officers	Ability to compare predictions with observations and official forecasts	Retain provenance, model version and verification history
Airport operations managers	Situational awareness and indication of potentially disruptive conditions	Provide summary views without presenting ML outputs as operational clearance
System administrators	Reliable ingestion, traceable errors and maintainable services	Log failures, validate data and document recovery procedures

1.3 Alignment with Sustainable Development Goals

The topic assessment form indicates that the connection of this project lies in the following sustainable development goals, such as SDG9: Industry, Innovation and Infrastructure, SDG11: Sustainable Cities and Communities and SDG13: Climate Action. The contribution of the project is indirect, as through research prototype, innovation can be achieved in local forecasting and assessment of weather data. There were no delays, fuel saving or less pollution or safety incidents measured.

1.4 Purpose and scope of the common report

This report provides an overview of the integrated research question, system architecture, shared workflows, interfaces between components, group-based evaluation strategy and project constraints. Rather than rehash all of the information concerning model development provided in the four separate theses, the report

describes each component at a level appropriate for understanding its place within the overall system, along with the data available for its integration.

1.5 Team members and component responsibilities

Table 1.2.Team members and component respon.sibilities

Team member	Student ID	Component responsibility	Primary output
Sankadeepa W.D.I	IT22113986	Temperature–pressure prediction and density altitude	Temperature/pressure forecasts and derived density-altitude indicator
Ranasingha R.A.A.D	IT22269720	Wind prediction, runway-oriented wind analysis and pilot portal	Wind forecast, headwind/crosswind interpretation, pilot-facing workflow
Jayawardhana H.M.S.I	IT22619976	Cloud and visibility prediction	Cloud/visibility prediction outputs and related interface pathway
Jayathilaka P.K.V.K	IT22069504	QNH, RH and dewpoint prediction	QNH and dewpoint estimates; RH derived from predicted dewpoint and temperature

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2. OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

Commonly, the problem lies in the lack of a complete and traceable end-to-end workflow at the airport level which would incorporate the generation of short-range forecasts of multiple meteorological variables related to aviation together with the methodology of their evaluation using observations. The problem does not merely lie in the lack of a specific machine-learning algorithm. It involves aspects like data quality issues, time synchronization, target inconsistency, component interoperability, and possible misinterpretation of the forecast outputs as operational aviation products.

The assessment of the provided topic uses T+3h as the forecasting period and suggests the comparison of the forecasts generated by it to official aviation forecasts and observations. It should be noted that the differentiation of the two terms is important. Comparison with a TAF and verification against a METAR observation are not identical operations, although they are linked. While TAF is the forecast, METAR is the observation.

2.2 Root causes and overall research gap

- Changes in airport-level conditions can occur faster and more localized than any regional prediction will show.

- Variables describing different weather phenomena have different mathematical formulations: wind direction is cyclical, clouds are categorical variables, and pressure, temperature, dew point, and wind speed are numerical.

- Historical data can include gaps, duplicates, wrong units or codes; temporal feature engineering decreases the number of rows usable.

- Components models can be developed using different train-test splits, time periods, target, and metrics, making direct comparisons between different components hard.

- Prediction records are not usable for verifying forecasts without the ability to match the time at which the prediction was made, prediction horizon, model ID, and observation.

- Presentation of operational predictions involves additional human factors.

This research gap identified by the topic assessment form is that of the hyper-local aviation weather forecasting system with verification methods that will be based on airport data with emphasis on monsoon effects. This problem is expanded into the common report question as to how various predictors can be integrated while still retaining their unique levels of evidential support.

2.3 Four subproblems and component mapping

Table 3.1.Overall problem and component mapping

Component	Subproblem	Approach documented in individual report	Integrated role
Wind and pilot portal	Forecast wind magnitude/direction and translate it into runway-relative quantities	XGBoost and TFT pathways; cyclic direction representation; deterministic headwind/crosswind calculations	Provides wind-specific outputs and a pilot-oriented access path
QNH, RH and dewpoint	Estimate pressure and moisture-related variables consistently	LightGBM for QNH and dewpoint; RH derived from predicted dewpoint and predicted temperature using August–Roche–Magnus relationship	Adds pressure and moisture context

Component	Subproblem	Approach documented in individual report	Integrated role
Temperature–pressure and density altitude	Forecast temperature and pressure and derive an aviation-oriented indicator	Reported RF–LSTM hybrid approach; density altitude derived from pressure altitude and ISA relationships	Provides temperature/pressure estimates and derived density altitude
Cloud and visibility	Estimate cloud and visibility conditions from airport weather records	Group paper reports Decision Tree, Random Forest and XGBoost comparison; implementation evidence requires reconciliation	Adds visual-weather context

2.4 Overall aim and group objectives

2.5 Scope, assumptions and constraints

However, the scope of the common case is restricted to the VCBI-based research prototype discussed in the materials provided. In the historic database presented in the material, there are 121,407 observations during the period of 2019–2026. As per the material provided, the three hours pipeline is expected to contain 51,175 observations after preprocessing and time feature construction. These numbers pertain to different stages and cannot be treated interchangeably.

The system works on the premise that the source data will be accessible or loadable with meaningful timestamps, the target values will have consistent interpretation, and each model will get its required features at inference time. The prototype is limited by the number of records available, the unequal evidence for the four elements, and the lack of documentation about formal validation from stakeholders and aviation

certifications. This is not to suggest that this prototype takes the place of METAR, TAF, expert forecasting, or flight operation practices.

2.6 Research questions

1. Could past observations from airports help make reliable three-hour-ahead predictions for various weather variables relevant to aviation at VCBI?
2. How would one combine the various component models and target representations without losing variable-specific units, target times, or provenance of evidence?
3. How would one transform component model predictions into aviation-related variables such as headwind, crosswind, and density altitude?

3. OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The recommended approach is an architecture of components rather than four separate applications. The common input comes from historical and fresh aviation weather observations. A validation and parsing process prepares fields and times. After that, particular feature creators prepare the input for each model. Predictions are saved with target time and information about a model. The common service layer provides access to values by a dashboard or pilot portal, and verification associates predictions with observations to compute error metrics.

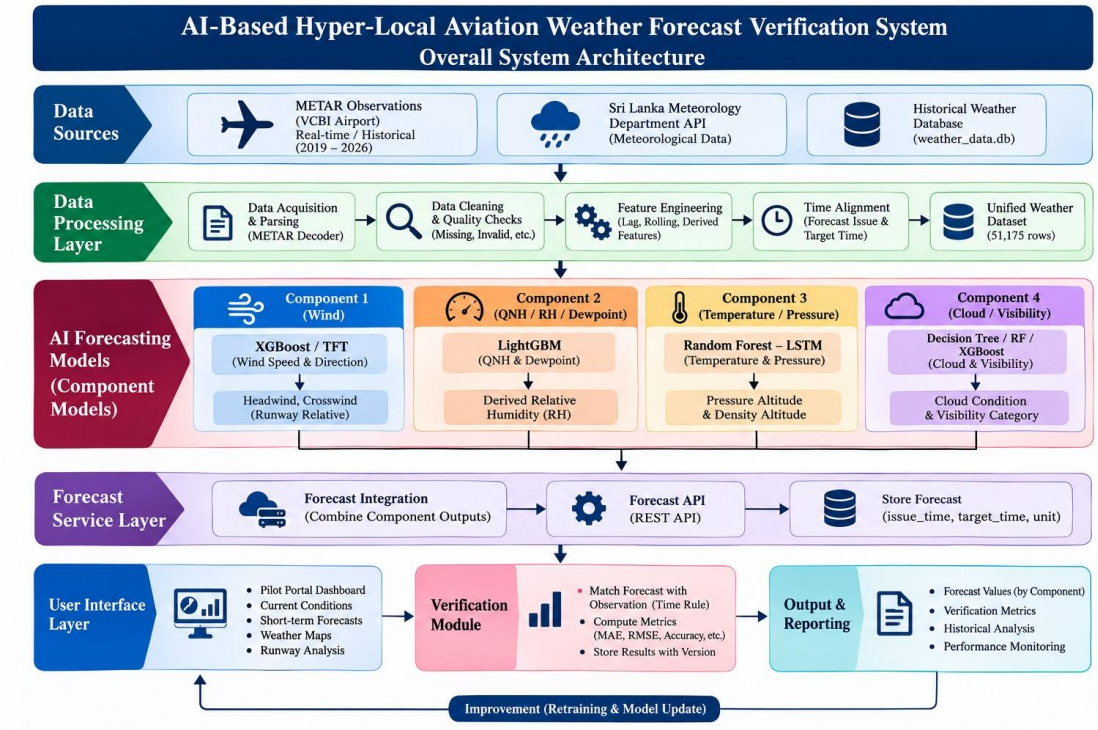


Figure 3.1: Integrated hyper-local aviation weather system

3.2 High-level architecture

The split of the design into the infrastructure that is shared and the per-component modeling ensures that a component can be modified without the need to modify all other predictors, provided that there is no modification to the interface between the inputs and outputs. It also presents the chance to test the components independently but using a shared persistence format.

The common layer shall treat the forecast as a formal data structure and not just a numerical value. The data structure shall contain, among other things, the following elements: the variable, the forecast value, its unit, the timestamp of issuing the forecast, the valid timestamp, the horizon, and the model name and version used for the forecasting. In any other case, one cannot tell whether the forecast is new or outdated.

3.3 Component 1 summary: wind and pilot portal

The wind component performs wind speed and wind direction prediction along with the conversion of the predicted vector into headwind and crosswind components in respect to the runway. The individual thesis achieves the following score for wind speed prediction based on the use of XGBoost algorithm: MAE = 1.0516 knots, RMSE = 1.4404 knots, and $R^2 = 0.7798$ on the dataset consisting of 10,235 test samples. Wind direction score is evaluated by using circular angular MAE score of 10.61 degrees. With respect to runway 04, the individual headwind MAE/RMSE score is 0.8833/1.2025 knots, while crosswind MAE/RMSE score is 1.2027/1.7270 knots. The wind component also includes pilot portal for flight planning along with weather briefing and aviation maps inclusion.

3.4 Component 2 summary: QNH, RH and dewpoint

The QNH / dewpoint part utilizes LightGBM models to predict the dewpoint and QNH in three hours. RH is calculated based on the predicted dewpoint and predicted dry-bulb temperature using the August–Roche–Magnus formula and not via an independent machine-learning RH model. As per the provided individual thesis, the scores of dewpoint and QNH are R^2 of about 0.66 and MAE of about 0.58 °C and 1.8 hPa, respectively. No error score for RH can be obtained from the provided individual thesis paper.

3.5 Component 3 summary: temperature, pressure and density altitude

The temperature-pressure model shows an example of a mixed approach, which includes both the Random Forest and LSTM models. Individual thesis assumes that there is an approximate R^2 of 0.84 for the mixed approach, 0.68 for the Random Forest model, and 0.25 for the LSTM model. The existing literature does not have sufficient data regarding the units and complete protocol of metrics measurement to make a comparison of these results with wind and QNH/dewpoint results. Density altitude is measured using the pressure altitude and ISA formulas. The density altitude value should be measured together with all input assumptions; it cannot be treated as the objective of the machine learning approach.

3.6 Component 4 summary: cloud and visibility

The fourth component involves prediction of clouds and visibility conditions through machine learning classification. The parameters mentioned above play an important role in assessing aviation weather and ensuring safety of flights. As it was noted in the reports, the accuracies of Decision Tree, Random Forest, and XGBoost are 76.25%, 76.31%, and 78.15%, respectively. The highest accuracy belongs to XGBoost.

At the same time, the mentioned results were not checked independently. It is necessary to verify the cloud categories, visibility units, targets, forecast periods and the discrepancy between the reported results for XGBoost and the script provided by the authors. It seems that the latter uses Random Forest classifier. Thus, the mentioned accuracies are retained as the results of the component report.

3.7 End-to-end user and data flow

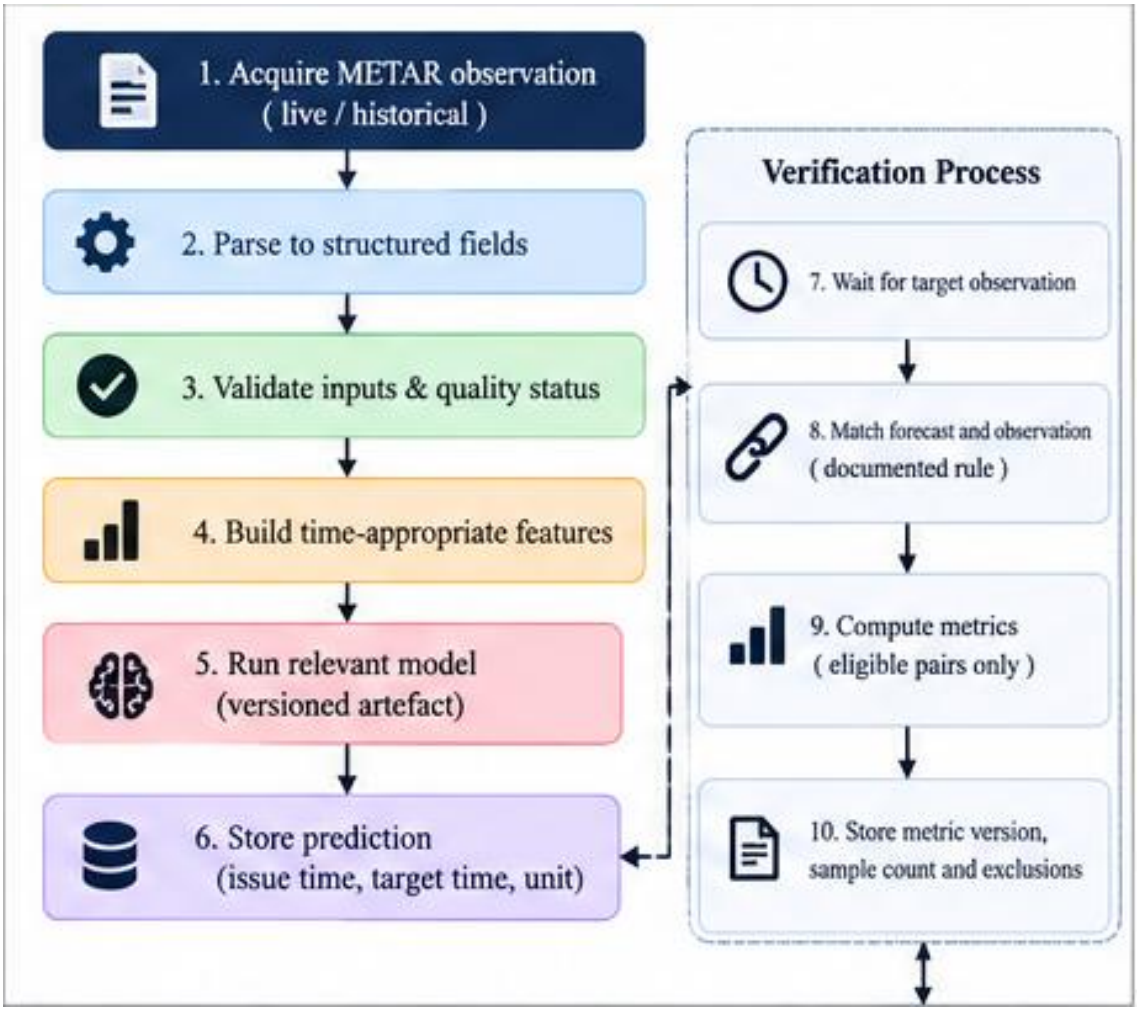


Figure 3.2: Forecast generation and verification workflow

To begin with, from a higher perspective, the process starts with an unprocessed observation, maintaining its origin. Next, the parsing stage transforms the report into fields. After that, validation performs a check on timestamps, units, mandatory values, and plausibility of the fields. Next is feature generation where the necessary temporal and meteorological features are generated for use in a component. Next is the inference process whereby predictions are made for the designated time of interest. Forecasts are stored in the process before being used by the interface. Once the target observation is available, there follows verification of the forecast.

3.8 Shared data and component boundaries

The four factors need not be fit into a single framework or a single performance measure. One framework can provide a standardized format for recording while letting each target variable keep its own measure. For example, wind direction needs circular error; wind speed and pressure need regression measures aware of units; cloud types may need classification measures; and density altitude needs equations laid out in the open.

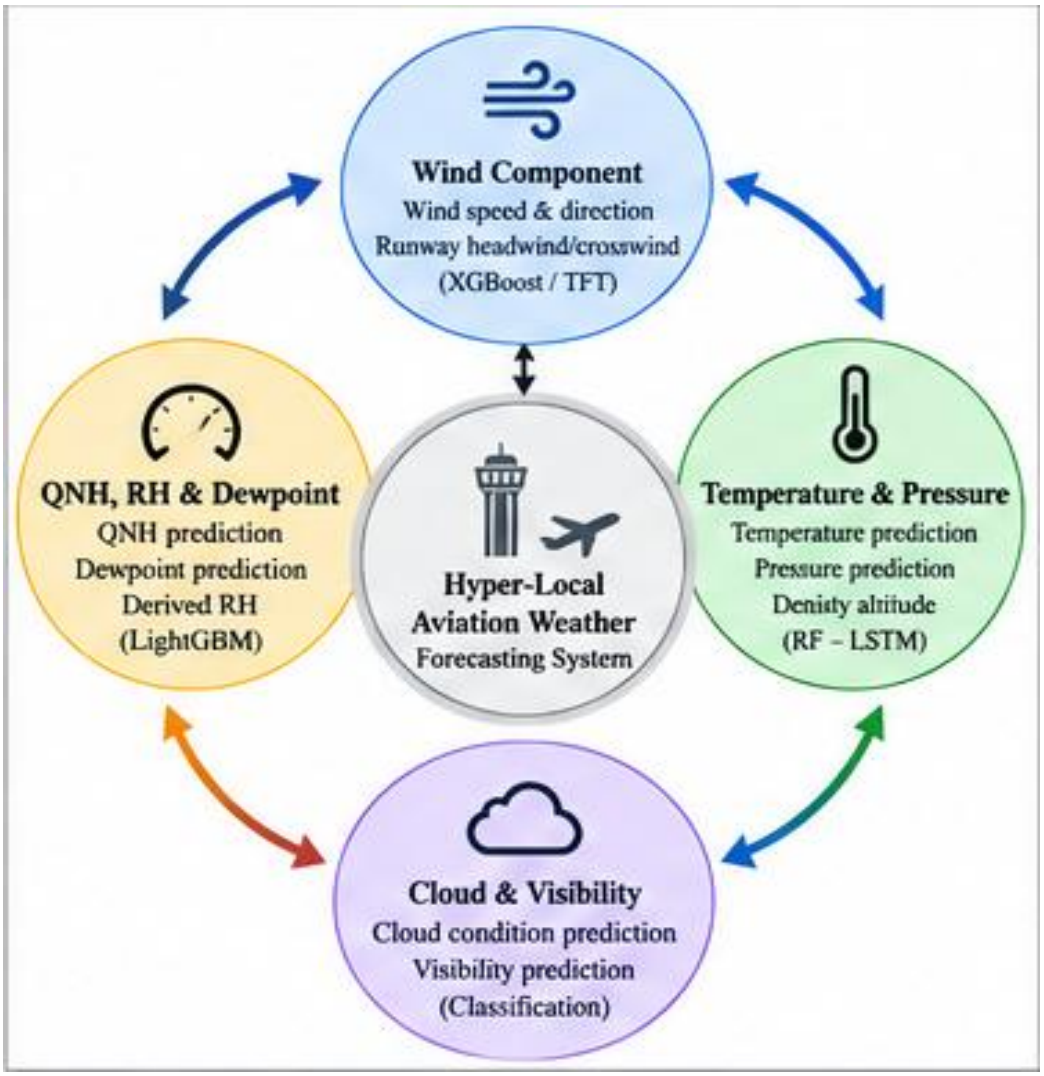


Figure 3.3: Four interdependent research components

4. COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

A sound approach to integration involves first coming to an agreement on the interfaces before linking the modules. Each module should offer the needed inputs, output variables, units, horizon of interest, and model meta-data. This should then be done in steps, where data is accessed, inferred from, persisted, presented, and validated. At each step, a limited number of known records should be used to ensure that values preserve their meaning through each step.

The project resources mention a common weather database and the modular back-end routers. The wind thesis speaks about the common forecast and verification architecture and SQLAlchemy data access. The cloud and visibility thesis discusses the process of METAR parsing, storing, Flask API, and dashboard. Meanwhile, the wind thesis speaks about the FastAPI process. These differences might be the result of different versions or different services implementation. The common thesis does not presuppose the deployment of each mentioned framework as a single runtime. The repository should reveal which services are running and how they interact.

4.2 Interfaces, data contracts and shared models

Common forecast data record must include, at least, unique record identifier, timestamp of the source observation, timestamp of the issued forecast, target timestamp, forecast lead time, variable name, predicted value, units, name and version of the model, version of the input data, and processing status. Verification data must be stored separately from prediction so that any corrections of the verification data do not overwrite the initial forecast data.

Table 4.1.Shared forecast data contract and validation rules

Contract field	Purpose	Validation rule
forecast_id	Unique forecast identity	Unique, non-null identifier
issued_at	Time forecast was generated	Valid timestamp and timezone convention documented
target_time	Time the forecast describes	Must be later than or equal to issue time; horizon derived and checked
horizon	Forecast lead time	Allowed values explicitly documented, including T+3 h
variable	Predicted meteorological quantity	Controlled vocabulary; no ambiguous aliases
value and unit	Numeric or categorical prediction	Unit compatible with variable and value type
model_id / version	Reproducibility	Must identify the artefact used for inference
quality_status	Communicate missing/stale/suspect inputs	Controlled status codes with defined handling
verification_status	Pending or completed comparison	Must not imply verification before target observation exists

4.3 Timestamp and temporal consistency

Temporal synchronization is one of the most critical integration issues. A forecast issued at a certain point and valid three hours later should be evaluated against an observation that represents the target time, not necessarily just against the following database entry. The reporting frequency is irregular, duplicates can exist in the observations, and the source may have a time stamp in some other timezone

convention. The integration process needs to specify what time standard it converts the time stamps into; UTC, time zone, or some other defined time zone.

In the provided wind thesis, the validation export has forecast-type labels like 1H, 3H, TFT_1H, TFT_3H, and HYBRID_3H, along with live data rows and duplicate timestamps. Such rows must not be automatically aggregated. In order to make an assessment reproducible, one must remove pending rows, deduplicate using a well-defined key, match the forecasts and observations according to their target times and forecast types, and recompute all metrics in a consistent way.

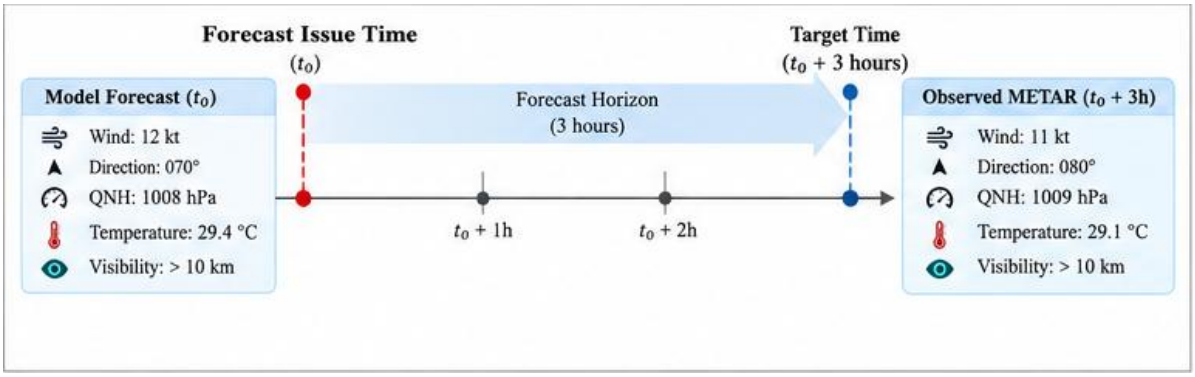


Figure 4.1: Time-aligned forecast verification

4.4 Units, target definitions and transformations

The handling of units needs to be clear in all interfaces for component interaction. The wind speed will be reported in knots for the wind thesis, wind direction will be in degrees, QNH will be in hectopascals, dew point and temperature will be in degrees Celsius, while visibility can either be in continuous distance or class representation. It is not proper for a system to just combine values due to similarity of the names of fields.

Transitions should be reversible. The wind direction should be encoded using the sine and cosine function and then decoded for display. The headwind and crosswind values can be expressed deterministically from the wind vector and the runway bearing values. RH in the QNH value is computed using predicted dew point and predicted temperature. Density altitude is computed using the pressure altitude and ISA relationships.

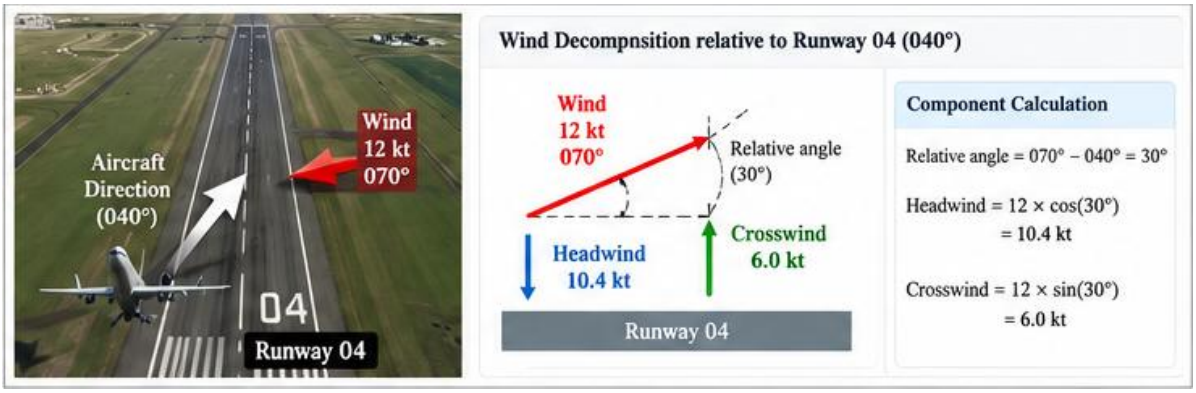


Figure 4.2Runway-oriented wind decomposition

4.5 Shared non-functional requirements

Table 4.2.Shared non-functional requirements

Requirement	Integrated interpretation	Evidence needed
Reliability	Do not return an apparently valid forecast when required inputs or models are unavailable	Error-path tests, logs and repeatable recovery checks
Performance	Inference and API response time should be measured under a stated workload	Timed tests with environment and sample size recorded
Usability	Users can distinguish observation, prediction, target time and advisory status	Task-based usability review; no formal study is claimed in supplied materials
Security	Protect pilot accounts and flight-plan information where the portal stores them	Authentication and access-control tests; secrets excluded from source control
Maintainability	Components can be updated behind stable interfaces	Versioned contracts, automated tests and deployment notes
Traceability	Every result can be linked to inputs, model version and target time	Stored metadata and reproducible evaluation records

Requirement	Integrated interpretation	Evidence needed
Responsible use	Outputs are supplementary and not operational clearance	Interface wording, model cards and limitations visible to users

4.6 Integrated workflow and exception handling

Normal operation involves the following steps: data collection, parsing, validation, feature creation, model prediction, storage, and presentation followed by verification. Exceptions may happen at any step in the process. Lack of availability of data source cannot be silently represented as a new observation. An error during parsing must be reported and skipped from model input unless there is a defined safe fallback approach. Errors in relation to missing model files or features mismatch must return an error message rather than fabricating values. If the target observation is not yet available, the forecast needs to wait for verification. In case of duplicated observations, selection must follow a rule.

Table 4.3.Workflow exception handling requirements

Failure condition	Required behavior	Audit record
Source/API unavailable	Retain last successful data only with a clear stale-data status; do not relabel it as current	Source status, failure time, last successful fetch
Malformed METAR	Reject or quarantine the record and preserve the parsing error	Raw record reference and parser error
Missing required feature	Do not run inference unless a tested fallback is defined	Missing feature names and component
Model version mismatch	Fail closed and report incompatible artefact	Expected/actual model and feature versions
No target observation yet	Keep verification status pending	Forecast ID and target time

Failure condition	Required behavior	Audit record
Duplicate forecast/observation	Apply documented deduplication rule before metrics	Rule version and discarded record IDs
Invalid or implausible value	Flag for review and prevent silent downstream use	Variable, value, threshold and decision

4.7 Team contribution and dependency matrix

Table 4.4.Team contribution and dependency matrix

Member/component	Research and modelling	Software contribution	Dependencies
Ranasigha R.A.A.D — wind and portal	Wind-speed/direction evaluation; runway transformations	Wind pipeline, backend integration and pilot portal described in individual thesis	Shared observations, forecast persistence, flight/map data sources
Jayathilaka P.K.V.K — QNH/RH/dewpoint	LightGBM for QNH/dewpoint; formula-derived RH	Feature pipeline and component inference described in individual thesis	Predicted temperature required for RH derivation; shared time-aligned observations
Sankadeepa W.D.I — temperature/pressure	Reported RF–LSTM hybrid and density-altitude calculation	Component data preparation and inference workflow described in individual thesis	Shared observations, pressure/temperature conventions and formula inputs
Jayawardhana H.M.S.I — cloud/visibility	Reported comparison of Decision Tree, RF and XGBoost	METAR parsing and cloud/visibility API/interface pathway described in individual thesis	Correct target representation, shared parser/data storage and dashboard contract

Member/component	Research and modelling	Software contribution	Dependencies
All members	Common problem, interface design and group evaluation	Cross-component integration, test evidence and report preparation	Version control, consistent definitions and evidence traceability

4.8 Integration risks and mitigations

First of all, the major integration risk is more semantic rather than technical: a value can pass through an API while pointing at the wrong valid time/unit/target. In order to mitigate this risk, it is necessary to validate the contract and traces but not just check that the service has returned HTTP success. The next risk is an inconsistency of evidence: the results of the same test can be produced on different splits but shown as comparable. In order to mitigate this risk, it is important to publish the component-specific protocol and use the common evaluation set only for the group claims.

5. INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The stated individual theses are indicative of a Python-based research stack. Pandas and NumPy are employed in the Python-based research stack for data management. In the Python-based research stack Scikit-learn is used for evaluation libraries. In the Python-based research stack, XGBoost and LightGBM are used for tree-based learning algorithms. In the Python-based research stack, PyTorch Forecasting is used for Temporal Fusion Transformer architecture. In the Python-based research stack, SQL is used for persistence. The wind thesis describes FastAPI, SQLAlchemy, SQLite, APScheduler, React/Vite and ReportLab, in the implementation context of the wind thesis. The cloud-and-visibility thesis describes Flask, in the pathway of the cloud-and-visibility thesis. Since these descriptions do not indicate a production deployment, the common report addresses the technologies described throughout the individual component artefacts.

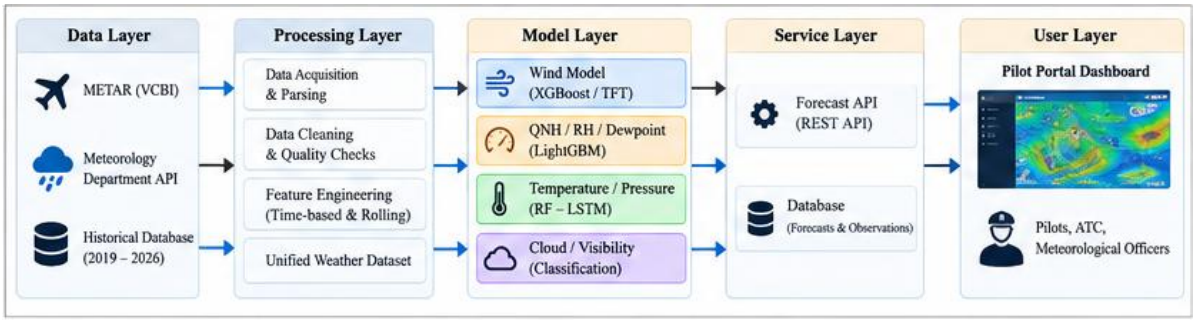


Figure 2.5:Common data-to-service architecture

Table 5.1.Common technology stack and integration considerations

Technology	Reported purpose in project materials	Integration consideration
Python	Data processing, model training and backend logic	Pin package versions and share environment instructions
Pandas / NumPy	Cleaning, feature engineering and vector calculations	Keep feature names and missing-value policies consistent
Scikit-learn	Metrics, model utilities and baseline comparisons	Use one metric implementation for final reported results
XGBoost	Wind prediction; reported cloud/visibility candidate	Store model version and exact target/feature contract
LightGBM	QNH and dewpoint prediction	Document training split and output units
PyTorch Forecasting / TFT	Temporal sequence forecasting pathway	Preserve chronological order and sequence metadata
Random Forest / LSTM	Reported temperature–pressure hybrid approach	Reconcile metrics and inference artefacts with experiment logs
SQLite / SQLAlchemy	Structured local storage and database access in wind component	Confirm shared schema and migration/version strategy

Technology	Reported purpose in project materials	Integration consideration
FastAPI / Flask	API service patterns described across individual reports	Clarify actual active services and API ownership
React / Vite	Dashboard and pilot-facing interface described in wind thesis	Display issue time, target time, units and limitations
APScheduler	Scheduled acquisition/maintenance jobs described in wind thesis	Prevent overlapping jobs and log failed runs
ReportLab	Generation of weather briefing PDFs described in wind thesis	Ensure generated documents identify forecast issue/target times

5.2 Data acquisition and preparation

The assessment form highlights the weather-data collection module, which captures real-time VCBI METAR data and determines the parameters. The wind thesis maintains a database of 121,407 observations between 2019 and 2026. The thesis states that the unified three-hour pipeline has 51,175 observations after the weather-data collection module deals with features and missing values. The figures serve as anchor points for dataset lineage; however, the submission will have to include a query or processing log to show how the figures were derived.

For a reproducible data preparation pipeline, it is important to ensure that the data is transformed from the raw format into structured fields, normalization of timestamps is done, all necessary variables are present, duplicates are handled, and there are any missing values; and all of this should be done before creating any features. All observations that are dropped should have a rationale behind such a move, whether there is a missing target, missing value, or missing lag/rolling window.

5.3 Feature engineering and leakage control

The person wind thesis involves recurring wind direction, dew point difference, recurring time of day features wind delay, moving average, QNH. QNH change. The

QNH/Dewpoint thesis involves time and moving features. The cloud/visibility report relates to interactions involving dew point difference, temperature, and RH, wind and RH, QNH and wind RH squared, and current weather coding. These features may be able to demonstrate temporal consistency and linear relations among them, but they require interpretation separately from each other per target instead of being interpreted consistently across all models.

This would imply that the feature is capable of only using information which is available at the time of prediction. In the case of moving averages, no data can be used after the prediction. Any transformation in which the scaling or grouping is learned through training data needs to be re-applied on the validation set, test set, and prediction. The time series modeling needs to preserve the order of data since any random shuffling of order may be appropriate only if the testing process allows for this.

5.4 Component model development

5.4.1 Wind

Wind Model Pathway utilizes XGBoost to predict in the table format. In addition, the Temporal Fusion Transformer Pathway is documented here. For the XGBoost model result, the split ratio is 80% / 20%, where random_state=42, consisting of 40,940 training samples and 10,235 test samples based on the documentated pipeline. Since a random split may put observations from seasons in both the data split, therefore the result should be viewed as the evaluation protocol and not proof of future-season generalization.

5.4.2 QNH, dewpoint and relative humidity

In case of the QNH and dewpoint prediction of the system, LightGBM algorithm is applied to forecast QNH and dewpoint in a three-hour window of time. Relative humidity is computed by using the predicted value of dewpoint and the predicted value of temperature according to the August-Roche-Magnus formula. Another score is required for the humidity prediction part but it is not mentioned in the above-mentioned paper. Therefore, the final version of the system will need to account for

the formula, the constants used, the units of input data and how to deal with unphysical outputs.

5.4.3 Temperature, pressure and density altitude

In the case of Sankadeepa W.D.I's component, there is an inclusion of temperature and atmospheric pressure predictions along with density altitude computation. In this individual thesis, the model used for prediction of all these parameters three hours ahead is the combination of Random Forest (RF) and Long Short-Term Memory (LSTM).

The hybrid approach gives a value of R^2 of 0.84 as opposed to RF, which gives an R^2 of about 0.68 and LSTM, which gives an R^2 of about 0.25. The hybrid MAE is 0.866. All values are taken from the source as it is presented. This should not be regarded as totally replicable because there is no information on the units of the target variable and sample size used. The reason why separate scores for temperature and pressure are given in the source despite the fact that they were not actually provided remains unclear. The component does not give the actual temperature and pressure data. It also gives pressure altitude and density altitude based on the forecast pressure and the ISA temperature model.

ISA temperature is the expected temperature at that pressure altitude. Deviation from ISA demonstrates the deviation of the forecast temperature from the standard. Finally, density altitude is obtained by taking into account the pressure and temperature for aviation purposes. The thesis uses a classification into Normal, Elevated and Critical conditions, depending on thresholds. Thresholds must match exactly to the code used, and cannot be used as universal standards for aviation. The evaluation of the components requires two phases. Firstly, it is necessary to compare the predicted temperature and pressure to the corresponding measured temperatures and pressures at the same forecast times, using MAE and RMSE, including the units for each of them. Secondly, calculate the density altitude based on the predicted input. Then compare it with the density altitude calculated based on the observed input, using the same formula. This will ensure that the chain of predicting and calculating the density altitude was validated. Unit tests should validate pressure altitude, ISA temperature

and density altitude calculation. The presented material allows to describe the method and the achieved hybrid results.

5.4.4 Cloud and visibility

The portion concerning the cloud and visibility prediction greatly depends on obtaining the METAR data and identifying the prediction targets. Treatment of visibility could either be in a quantitative or categorical manner. In the case of the clouds, possible attributes include the degree of cloud cover, the height of the lowest cloud layer, or an indicator showing the overall cloudiness condition. The metrics that are used to evaluate the performance of the model should correspond to the target definitions. The accuracy figures that are provided should go along with the number of instances as well as how the classes were defined, their distribution, and the confusion matrix if classification was used. Individual thesis identifies a discrepancy between the XGBoost output of the paper and the output of RandomForestClassifier coding; such discrepancy should be corrected by using the code and the evaluation results.

5.5 API, persistence and interface integration

The service layer should display the predictions along with the target time and additional information. Pilot portal which is stated in the wind thesis applies flight information to build a briefing which provides aviation map products and TAF data to produce a briefing PDF and store a flight plan record. It is one of the system components as it displays the output of the models as per user workflow. Simply displaying the screenshot of the user portal will not prove that all the model outputs are right. Proof in solid terms will be in form of the flow from METAR data, feature vector fields, prediction, API response and finally on-screen output.

5.6 Deployment and operational configuration

Wind thesis involves discussing a system that operates continuously with scheduled jobs. A good design should ensure that the following elements are included in the configuration for security and functionality; keeping code settings secure, keeping password secure, keeping track of all scheduled jobs, preventing simultaneous

scheduling, checking of all parts of the model during the initialization process of the system and indicating proper functionality. The backups must include not only the input data but also the output results and checks. It is better to test the system in case of recovery rather than hope that everything would work out. Every change of the model must include versioning and the operational model must remain accessible until the tests of the new version are passed.

5.7 Evidence of team execution

The commonly used guidelines for the report call for a discussion of evidence for the implementation of the project from planning tools, communications, Git commit meetings, and research diaries. The provided files determine the sub-topics of the components. Model results are reported but complete task-board exports, commit history, meeting minutes, and signed progress logs are not included. Thus, the thesis does not make up sprint dates. Assert that a particular CI pipeline has been utilized. The final team must include an evidence table with actual repository commits, meeting minutes, and task artifacts if these are needed for evaluation.

Table 5.2.Implementation evidence categories and status

Evidence category	What should be attached or referenced	Current status in supplied materials
Source repository	Commit hash, branch and tagged release used for final build	Not included in supplied common-report file
Experiment logs	Training configuration, split indices, metric output and saved predictions	Partial metrics are present; complete logs not supplied
Integration tests	Request/response examples and end-to-end trace	Recommended; formal integrated test report not supplied
Team execution	Task board, meeting records or research diary extracts	Not supplied in current document set

Evidence category	What should be attached or referenced	Current status in supplied materials
Deployment	Environment file, deployment steps, backup/restore record	Descriptions exist in individual reports; operational proof incomplete

6. INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The evaluation framework considers the following four questions: Whether the prediction of components is useful in terms of figures based on its testing data, Whether the data and the modification of the data is done properly, Whether the output of the component passes the interconnected components without any change in meaning, and Whether the users of the system understand the output without any problem. However, scoring good in one aspect does not necessarily imply that others have been performed well.

In regression, the MAE reveals the mean difference between the values while the RMSE gives greater weight to larger differences. The R^2 metric compares the magnitude of the errors with the variation in the target values within the selected test data set. When dealing with wind direction, the error metric should consider the 0° and 360° boundary. In classification, accuracy should be supplemented by per-class metrics such as precision, recall, F1-score and confusion matrix, particularly in cases where danger scenarios are rare. Generated predictions should be evaluated independently of the predictive model on their own terms.

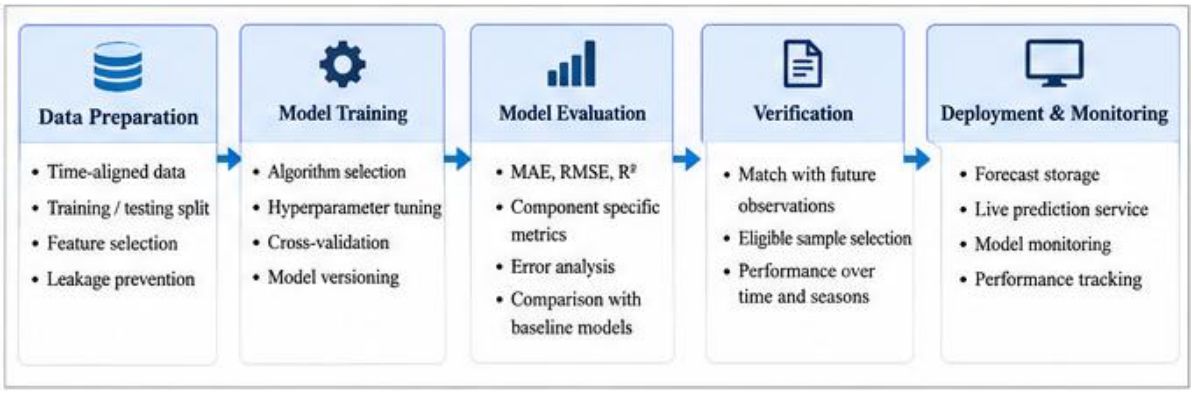


Figure 6.1.Evidence-led model evaluation workflow

6.2 Reported component-level results

Table 6.1.Reported component-level model results

Component / output	Reported result in supplied individual materials	Evidence limitation
Wind speed, XGBoost	MAE 1.0516 kt; RMSE 1.4404 kt; R^2 0.7798; n=10,235	Random 80/20 split with random_state=42; should add chronological holdout
Wind direction	Angular MAE 10.61°	Direction-specific sample count and subgroup results should be confirmed
Runway 04 headwind	MAE 0.8833 kt; RMSE 1.2025 kt	Derived value; runway heading and vector convention should be recorded
Runway 04 crosswind	MAE 1.2027 kt; RMSE 1.7270 kt	Derived value; threshold outcomes and operational interpretation require care
Dewpoint, LightGBM	MAE ≈ 0.58 °C; R^2 ≈ 0.66	Test sample count, split boundaries and full residual distribution not supplied
QNH, LightGBM	MAE ≈ 1.8 hPa	QNH R^2 and RMSE not reported in supplied paper
Relative humidity	Derived from predicted dewpoint and predicted temperature	No separate RH error score provided; evaluate on paired observations
Temperature–pressure hybrid	Reported R^2 ≈ 0.84; RF ≈ 0.68; LSTM ≈ 0.25	Metric target/units and full evaluation protocol need confirmation
Cloud / visibility	Decision Tree 76.25%; RF 76.31%; XGBoost 78.15% reported accuracy	XGBoost implementation/evaluation discrepancy and target/metric definitions need resolution

The accuracy results do not get combined into an average for a reason. The accuracy is dependent on variables, units and metrics. There are also model families and perhaps even varying data partitions used. An average result which combines knots, degrees Celsius, hectopascals, angular degrees and percentage classification results wouldn't mean anything. In fact, such an average would have no meaning at all. What needs to happen is that proper evaluation gives its own separate accuracy value for every individual metric. It should also give system-wide metrics like timestamp match rate, parser success rate, API success rate, forecast coverage and end-to-end traceability.

6.3 Interpretation of wind results

The MAE of 1.0516 knots for the wind speed is nearly one knot away from one knot in the testing set. The RMSE value of 1.4404 knots indicates that large errors affect the squared error more than the MAE. The R^2 value of 0.7798 suggests that the fitted model captures some portion of the variance in the target. All these statements are just a description of the reported results. However, they cannot guarantee good performance under all phases of the monsoon and high-impact situations.

MAE of 10.61 degrees is angular MAE that is used to evaluate wind direction instead of using simple absolute difference. So, the difference between wind direction of 359° and 1° is 2°, not 358°. The use of runway-based criteria provides the additional level of analysis that converts vector forecast into runway coordinate system. This is deterministic process if we have wind speed, wind direction and runway orientation known. Nevertheless, it is influenced by forecast errors and chosen runway arrangement.

6.4 Interpretation of QNH, dewpoint and RH results

The dewpoint measurement of around 0.58 °C MAE and 0.66 R^2 is an example of the assessment provided for the dewpoint. The QNH MAE of around 1.8 hPa is the pressure measurement result. It is necessary to take note of the absence of the RH result as the RH is a derived measurement that depends on the predicted dewpoint and temperature results. Errors in both of the input parameters could affect the RH value. The final team assessment should include the RH errors relative to the actual RH value at the target time.

6.5 Interpretation of temperature–pressure and density-altitude results

The R^2 of the temperature-pressure thesis is around 0.84, RF result is around 0.68, and LSTM result is around 0.25. The hybrid MAE is 0.866. All these figures have been cited directly from the thesis without any third party's recalculations. In order to make sure that the hybrid has indeed performed better than other two methods in the thesis evaluation, we should make sure that all three models have been applied to the same targets, the same test cases, the same units, and the same method of calculating the metric. Neither the summary of the thesis nor the work of Sankadeepa presents any figure related to the MAE or RMSE for the specific targets (temperature and pressure). Thus, there is no information about the combination of these two metrics as well as the unit used. The work of Sankadeepa also focuses on pressure-altitude and density-altitude. Density altitude is a metric used in aviation; it is not a machine learning target in the workflow discussed here.

Thus, the analysis should begin by comparing the calculated temperature and pressure with the actual values of these variables. Then, the calculated density-altitude from temperature and pressure should be compared with a reference density altitude calculated using the temperature and pressure, using the same equation. The final report should mention the number of samples used, units of measurement used, the version of the equation used and the method for dealing with missing and erroneous data. This source provides details about the calculation process but there is no validation of density-altitude error and therefore no score can be obtained.

6.6 Interpretation of cloud and visibility results

The accuracy values rise from 76.25 percent in Decision Tree to 76.31 percent in Random Forest and 78.15 percent in XGBoost. Comparing all three algorithms on the data sets having the same target classes and the same criterion for measurement will show that XGBoost outperforms the other two models. The individual thesis proves that there is a gap between the results of XGBoost in the paper and the code. The team cannot claim that the findings are replicable until the code, the model, the predictions and the results of the evaluation are aligned. In case when the value to predict is

numeric, accuracy is not an adequate measure of performance unless the number is categorized intentionally.

6.7 Integration and end-to-end testing

Table 6.2.Integration and verification test catalogue

Test ID	Test objective	Expected outcome	Evidence to record
INT-01	Ingest valid METAR	Structured record created with source timestamp	Input sample, parsed fields, status
INT-02	Parse malformed METAR	Record rejected or quarantined without corrupting pipeline	Raw input and error log
INT-03	Validate timestamps	Timezone and target-time convention applied consistently	Before/after timestamp example
INT-04	Build features	Required features created without future leakage	Feature names and representative values
INT-05	Load model	Correct model/version loaded with expected features	Model metadata and startup log
INT-06	Generate T+3 h prediction	Output includes issue time, target time, unit and model ID	API request/response pair
INT-07	Store forecast	Forecast saved without overwriting source observation	Database row and unique identifier
INT-08	Display prediction	Interface labels forecast status and target time clearly	Screenshot plus API trace
INT-09	Verify forecast	Prediction paired with actual observation at target time	Pairing record and calculated error

Test ID	Test objective	Expected outcome	Evidence to record
INT-10	Handle pending target	Forecast remains pending until observation exists	Status record
INT-11	Handle stale source	Interface flags stale data or suppresses forecast	Failure simulation and UI state
INT-12	Derived wind values	Known vector inputs produce expected headwind/crosswind	Unit test input/output
INT-13	Derived RH	Known temperature/dewpoint values produce expected RH within defined tolerance	Formula test and observed comparison
INT-14	Density altitude	Documented formula yields reproducible value for known inputs	Calculation trace and units
INT-15	Duplicate handling	Deduplication rule prevents duplicate scoring	Counts before/after and rule version

A test catalog is presented in the above table. This does not mean that all tests have been tested and that the results were positive. In the final test results document, each test must be clearly stated to be a pass, fail, blocked or not run. Evidence should be provided for each test. One cannot assume any test result based on a design description.

6.8 Stakeholder and user validation

Since the report provided is an individual report, it lacks a stakeholder interview, a usability test, and an acceptance test. Therefore, the common report cannot state whether pilots and meteorologists approved the design. In future testing, the users will be required to indicate the target time to differentiate between observation and prediction; to detect when there are old or no data; to understand the relative runway information provided and to explain the fact that the tool does not provide operation

clearance. The report will have to include the tasks, roles, questions, findings and limitations, but not the personal information of the users.

6.9 Evidence gaps and required confirmation

Table 6.3.Evidence limitations and items requiring confirmation

Area	What is supported	What must be confirmed before final submission
Dataset lineage	121,407 database records; 51,175 rows reported after unified 3H pipeline	Exact database version, query, date coverage and stage-by-stage row counts
Wind metrics	MAE/RMSE/R^2 and angular/runway metrics are reported	Final prediction CSV, split indices, duplicate handling and season-wise metrics
QNH/dewpoint	Approximate MAE/R^2 figures are reported	Exact test sample count, boundaries, units and full metric output
RH	Formula-derived using predicted dewpoint and temperature	Observed-RH validation, formula constants and handling of out-of-range values
Temperature–pressure	Hybrid RF–LSTM and approximate R^2 figures are reported	Target-specific MAE/RMSE, target units, combination rule and experiment logs
Cloud/visibility	Reported accuracy comparison is present	Reconcile XGBoost result with implementation script; define labels and error metric
Integration	Modular architecture and service pathways are described	Actual active service topology, API contracts and end-to-end test traces
User validation	Stakeholder groups are identified	Formal validation evidence, if it occurred; otherwise mark as future work

Area	What is supported	What must be confirmed before final submission
Deployment	Scheduled operation and portal functionality are described	Deployment logs, health checks, backup/restore and security test evidence

6.10 Threats to validity

The following factors could undermine internal validity: data leakage, data duplications, inconsistency of features construction or the differences between training data and actual use of data. Target definition could undermine construct validity, particularly in regard to cloud coverage and visibility. External validity is low since the research covers just one airport, and we need analyses for the monsoon season and for rare events. Statistical conclusions could be questionable due to lack of sample sizes, confidence intervals and residual distribution. Integration validity is low since actual integration process has not been revealed yet.

The challenges do not render the project useless, but rather reveal when the potential research prototype ends and when the system requires proof in order to become adopted on a broader scale. It is much better to document the unknown than to provide data that lacks justification.

7. COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The suggested value proposition is about a multivariate forecast approach which is aimed at fitting the outputs of the models in the consistent workflow while retaining the capability of comparing those outputs with observations. The intended users would be pilots, aviation meteorologists, airport operations staff, and researchers analyzing the behavior of weather systems. The value of the system does not lie in the substitution of the forecasts but rather in the creation of the research environment for the short-range forecasting.

7.2 Market alternatives and differentiation

In the list of products provided in the topic assessment form, it is mentioned that METAR and TAF are available, along with the weather forecasting systems including Global Forecast System and ECMWF products, which are available in the current scenario. They are reliable and used on a daily basis, hence labeling them as old and unreliable is unjustified. The new study, however, differs from them in terms of its focus and approach to airport weather forecasting.

The unique aspect of the project approach remains to be proven until benchmarked against actual benchmarks within the same period. An appropriate comparison is done between the project approach and persistence. This refers to the assumption that the weather tomorrow will be similar to that of today. This is done using the statistical technique which becomes the official forecast product where there is a suitable match of the target and the machine learning models. Every test must specify the target, lead time, number of cases and the performance metric used.

7.3 Business model and feasibility

Possible future delivery models may include a research dashboard for weather data analysis, an internal decision support system, or an integration layer that facilitates understanding of forecast data.

There is no commercial deployment, commitment from the client, any financial gain, operating costs, or market demand surveys provided in the files.

Thus, there can be no claim about a validated business model in the report.

The feasibility study will require consideration of data license fees, infrastructure maintenance costs, monitoring of the model support responsibilities, security, government approval, and safety of information.

7.4 Adoption, intellectual property and compliance

Prior to adoption, there is the need for the project to sort out the licensing and ownership of code, data, and third-party library; document permissions for source-data; and look at the relevant institutional and aviation regulations. The models' artefacts ought to be versioned according to the training data range, targets, limitations, and use. Any deployment for use by users ought to state that the predictions are not METAR/TAF or flight clearance.

7.5 Responsible AI and safety

Responsibility in this context means measured communication, as opposed to overly confident communication. The interface must tell if a value is an observation, forecast, machine learning prediction, or some other kind of calculation. It should show when the issue was last seen and when the target will occur. It should alert the user if some input is missing or has gone bad. The user should not be led to feel more certain because there's only one number.

While the weather observations that are being used to train models are not personal by definition, it is noted that user and flight plan information is present in the pilot portal in question from the wind thesis. Authentication information and user records need to be protected using least privilege and secure storage, transport and retention of credentials. This document does not prove a completed security assessment.

7.6 Sustainability and Sustainable Development Goals

The Topic Assessment Sheet shows a potential link between the topic and SDG 9, 11, and 13. The best supported statement with evidence is that the project is working on

an innovative digital technology for local weather prediction in an attempt to promote safe decision-making. This project has not yet considered any impact on the emissions from aircraft, delays, or accidents.

7.7 Risk register

Table 7.1.Integrated project risk register

Risk	Potential effect	Mitigation / control	Evidence status
Incorrect forecast target time	Misleading comparison and interpretation	Enforce target-time contract and pairing tests	Control proposed; test evidence required
Poor performance during rare events	Important weather cases may be missed	Seasonal and event-based evaluation; report recall for hazard classes	Additional evaluation required
Model or feature mismatch	Invalid inference or silent output drift	Version model and feature schema together	Implementation check required
Source-data outage	Stale or missing forecasts	Health monitoring, explicit stale status and retry policy	Operational evidence required
Cloud/visibility metric ambiguity	Misleading model comparison	Confirm target representation and recalculate from saved predictions	Known unresolved evidence gap
User over-reliance	Prediction mistaken for operational clearance	Visible limitations and clear source/status labels	UI review required
Credential or flight-plan exposure	Privacy and account compromise	Least privilege, secure secrets and access testing	Security test evidence required

Risk	Potential effect	Mitigation / control	Evidence status
Model drift	Performance degrades over time	Monitor verified errors and use controlled retraining	Longitudinal monitoring not evidenced
Unsupported impact claims	Overstatement of safety or sustainability benefit	Report intended contribution separately from measured impact	Editorial control required

7.8 Scalability and future maintenance

However, scalability for the modular system design may apply to airports or variables, but it is more than just scaling of the database. For instance, new airports will require observation frequency, varying climatology, runway configuration, and varying patterns of missing data. All these new airports will be required to carry out data verification, model testing, and assess if the model developed using VCBI can generalize. The project needs to maintain a model registry with verified errors in respect to time and weather regimes.

8. CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The research has. Conducted an integrated methodology for hyper-local aviation weather forecasting and verification based on the VCBI airport. The methodology is comprised of four parts each targeting different weather parameters: wind and surface condition; QNH, dew point, and consequently, relative humidity; temperature and pressure for density altitude; cloudiness and visibility. These parts utilize models and methods, but they have the same architecture. The same architecture enables treating these parts as a system even though their modeling approaches, objects of analysis and verification requirements remain different.

It can be seen from the evidence provided above that the results of the modeling have been provided for each part separately. The part on the wind provides the detailed numerical information, including the description of the size of the sample used for the tests and some particular metrics related to the operation of runways. The QNH and dew point part provides the approximate values of errors, although the value of relative humidity is calculated based on the provided parameters. There is no particular score for this part of the report. The temperature-pressure mixed model provides the values of R^2, which indicate good prediction power of the model but there are no metrics related to particular objectives and there is a lack of information about the test procedure.

The major lessons learned from this study are twofold. The first lesson is that it is possible to connect forecasting methods into a unified process for forecasting and verifying using a modular system. The second lesson is that for this kind of modular system to function effectively, time synchronization, uniform units, sources of data, replicability of testing and communication of uncertainty need to be addressed. From the evidence presented, the system can only be classified as a research prototype since it is not certified for use in the aviation industry.

8.2 Principal integrated contribution

- Problem definition and architectural design for four interrelated airport weather forecasting modules.

- A standardized data description agreement in terms of forecasts, times to target, units, model versions and verified status.
- Documentation of the process from observations, feature engineering, inference, persistence, visualization to eventually verification.
- An evidence-based summary of the metrics per module retaining their units and limitations.
- A checklist of tasks to do prior to go into operations for integration, evaluation, safety and reproducibility.

8.3 Achievement of common objectives

Table 8.2.Common objectives and achievement summary

Objective	Evidence in supplied materials	Status
Define the common problem and gap	Topic Assessment Form R26-IT-132 and individual theses	Documented
Describe four components and their roles	Four individual thesis documents	Documented
Define an integrated architecture and data flow	Component architecture and workflow described in individual reports; integrated diagram included here	Conceptual design documented; implementation trace required
Report component model metrics	Reported results for wind, QNH/dewpoint, temperature-pressure and cloud/visibility	Partially achieved; several protocol details need confirmation
Define consistent forecast verification	Wind report identifies timestamp/duplicate/pending-row issues; this report defines matching protocol	Protocol documented; common test results not supplied

Objective	Evidence in supplied materials	Status
Demonstrate end-to-end integration	Individual reports describe backend and interface paths	Not fully evidenced in supplied materials
Validate with stakeholders	Stakeholder groups identified	Not evidenced; future work unless records exist
Establish operational readiness	Safety limitations and controls documented	Not achieved; prototype only

8.4 Future directions

1. Freeze a version of the data set and release the count at each cleaning and feature engineering stage.
2. Create a common export for evaluation which contains: issue time, target time, forecast horizon, model version, target value, and associated observation.
3. Conduct a chronological holdout evaluation in addition to the reported wind split, including seasonal and event-wise breakdowns.
4. Validate the cloud visibility XGBoost prediction against the source code and saved predictions; clearly define target and metric.
5. Report target-specific MAE and RMSE for temperature and pressure, while assessing density altitude separately from its predictive inputs.
6. Compare calculated RH with observed RH via time-synchronized predictions and document all formula constants and constraints.
7. Document integration test catalog, including parsing issues, data staleness, model feature gaps, API response and forecasts match.
8. Organize an organized stakeholder review centered around interpretation correctness, timeliness, data source differentiation and limitations.
9. Create model cards, dashboard for monitoring, drift detection and retraining process.
10. Consider data accessibility, deployment costs, security, licensing and regulatory/institutional approval if necessary.

8.5 Final statement

What makes this project significant is the attempt to relate forecasting of the weather to aviation-specific interpretation and validation. What should come after is not making unsubstantiated assertions or complicating the models. Rather, it should be addressing the current lack of evidence which prevents any evaluation of the outcomes at the component level.

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GLOSSARY

Term	Definition
Aviation weather	Meteorological information used in flight planning and aviation operations
Crosswind	Wind component perpendicular to the runway heading
Density altitude	Altitude in the standard atmosphere at which air density would equal the actual density
Dewpoint	Temperature to which air must be cooled, at constant pressure and water-vapour content, to reach saturation
Forecast issue time	Time at which the forecast was generated
Forecast target time	Time for which the predicted condition is intended to apply
Headwind	Wind component aligned against the aircraft's direction along the runway

Term	Definition
Hyper-local forecast	Forecast intended for a geographically specific location such as an airport terminal
MAE	Mean absolute error, the average magnitude of prediction error in target units
METAR	Routine aviation meteorological observation report
QNH	Aviation altimeter setting associated with pressure reduced to mean sea level
Relative humidity	Ratio, expressed as a percentage, of actual water-vapour pressure to saturation vapour pressure at the same temperature
RMSE	Root mean squared error, a measure that weights larger errors more strongly
T+3 h	A prediction target three hours after the forecast issue time
TAF	Terminal Aerodrome Forecast for an airport and a specified period
Verification	Comparison of a forecast with an observation corresponding to the forecast's target time

APPENDICES

The appendices are there to support with additional technical information the four inter-related research pieces: wind prediction and interpretation relative to the runway; QNH, dew point and derived relative humidity; temperature and pressure prediction with the calculation of density altitude; and cloud/visibility evaluation. The difference is made between the values given in the provided documents, the examples of implementation and those tests which still have to be done.

Appendix A: Data Collection and Preprocessing

A.1 Purpose and evidence boundary

This documentation of the project indicates the collection of VCBI METAR data as well as the use of a shared historical database from 2019 to 2026. Wind Individual Thesis has recorded a total of 121,407 raw database records and 51,175 database records that have been kept as a result of the three-hour pipeline used. This is a source record; thus, the exact count at each stage should be obtained from the database and pipeline logs.

A.2 Data collection and processing described in the project material

Table A.1: Data collection and preprocessing stages

Stage	Operation	Evidence to preserve	Main risk
1. Acquisition	Retrieve METAR / historical observations from the approved source.	Fetch timestamp, source identifier, raw report and request outcome.	Missing intervals or source outages.
2. Parsing	Convert raw report fields into structured variables.	Parser version, parsed fields and warnings.	Malformed or ambiguous codes.
3. Normalization	Standardize timestamps, units and column names.	Time zone rule, unit map and schema version.	Mixing local time and UTC.
4. Quality checks	Detect invalid values, duplicates and missing targets.	Row counts and reason codes for exclusions.	Silent row deletion.
5. Feature engineering	Create lag, rolling, cyclic and interaction features.	Feature definitions and availability time.	Future-data leakage.

6. Target alignment	Pair inputs at issue time with target at T+3 h.	Issue time, target time and matching rule.	Comparing the wrong observation.
7. Model-ready export	Create per-component train/validation/test sets.	Dataset hash, split rule and row counts.	Inconsistent component samples.

A.3 Documented preprocessing approach

These data sources provide information on the pipeline that analyzes and verifies weather observations, sorts data by timestamp, computes lags and rolling variables, and excludes observations which can't be utilized for the chosen predictive variable. It has been determined that the specific execution process was not derived from the general report and therefore cannot be substituted for any other code.

A.4 Leakage prevention and split controls

The lagging and rolling transformations should use only the observations available at the forecast issuance. The rolling window should not include the future target variable. The scaling, imputation, and encoding transformers that learn their parameters from the data should fit on the training data and apply without modification to the validation and test datasets. For sequential models, observations should preserve their temporal ordering and sequence bounds should not span across the train-test split.

Table A.2: Leakage prevention and data-splitting controls

Control	Implementation rule	Verification evidence
Chronology	Sort by observation timestamp before temporal features.	Sample rows before and after sorting.
Target horizon	Match each feature row to the intended T+3 h observation.	Issue-time/target-time pairs.
No future features	Rolling and lag calculations use past/current values only.	Unit test with a synthetic ordered sequence.
Train-only transforms	Fit data-dependent transforms on training data.	Saved pipeline object and split code.

A.6 Component-specific preprocessing considerations

Table A.3: Component-specific preprocessing considerations

Component	Feature / preprocessing considerations	Checks required
Wind	Wind direction is circular; cyclic encoding and runway-relative vector calculations may be used. The individual wind report mentions lags, rolling means, cyclic hour and direction features.	Direction wrap-around, units, runway heading and target timestamp.
QNH / dewpoint / RH	QNH and dewpoint are forecast by LightGBM in the supplied report; RH is derived from predicted dewpoint and temperature using August–Roche–Magnus.	Input units, same target time, plausible RH range and formula constants.
Temperature / pressure / density altitude	The supplied individual thesis describes an RF–LSTM hybrid for three-hour temperature and pressure prediction, followed by aviation-related transformations.	Sequence order, target-specific units, hybrid combination rule and independent density-altitude calculation.
Cloud / visibility	METAR cloud groups, visibility codes and present-weather encodings require explicit target definitions.	Parser behaviour for missing, variable, coded and malformed reports.

Table 0.A.4: Dataset documentation fields

Item	Value to record in final submission
Database file and version	Exact filename, date, size and checksum.
Raw record count	Reported as 121,407 in the wind individual thesis; confirm against final database.
Pipeline output count	Reported as 51,175 for the unified 3H pipeline; state the exact stage and rule.
Date range	2019–2026 as described in supplied project material.
Feature schema	One schema per component with units and target definition.
Code version	Repository URL, commit hash and environment file.
Execution log	Run date, command, warnings, exclusions and output paths.

Appendix B: Dataset Structure and Feature Descriptions

B.1 Shared record structure

While the general approach uses multiple meteorological variables, the data ready for modeling cannot be considered as one homogeneous table. Each feature can have its own selection of columns, number of samples, and split ratio. The following list of items is a template of documentation according to the variables described in the provided documents; however, the actual database column names and types have to be verified.

Table Shared dataset record structure

Field / group	Meaning	Typical role	Documentation requirement
observation_time	Time represented by the observation.	Index / alignment key.	Timezone, precision and duplicate rule.

METAR raw text	Original aviation weather report.	Source evidence.	Source and parser version.
wind_speed	Observed wind speed.	Wind target / feature.	Unit, calm-wind handling.
wind_direction	Observed wind direction.	Wind target / feature.	Degrees, variable-direction handling.
QNH	Altimeter setting / pressure field used by component.	QNH target / feature.	Exact source field and hPa convention.
temperature	Dry-bulb temperature.	Temperature target / feature.	Degrees Celsius and missing-value rule.
dewpoint	Dewpoint temperature.	Dewpoint target / RH input.	Degrees Celsius and physical checks.
relative_humidity	Humidity percentage.	Feature or derived output.	Observed or derived flag; range 0–100%.
visibility	Reported visibility distance or class.	Visibility target / feature.	Units, code parsing and class thresholds.
cloud groups	Coverage, cloud-base or vertical-visibility codes.	Cloud target / feature.	Target definition and parsing rules.
present_weather	Weather phenomena code.	Feature / class context.	Encoding dictionary and unknown class.
issue_time / target_time	Forecast creation and valid time.	Verification keys.	Target horizon and timestamp tolerance.

B.2 Feature groups

Table B.2: Feature groups and their purposes

Feature group	Examples reported in source material	Purpose	Caution
Lag features	Previous wind, QNH and other weather values.	Represent persistence and recent change.	Must be available at issue time.
Rolling statistics	Rolling mean over recent observations.	Smooth short-term variability.	Window size and min_periods must be recorded.
Cyclic time features	Hour-of-day or direction represented cyclically.	Avoid discontinuity at 0/360 or 23/0.	State exact formula and period.
Interaction features	Dewpoint depression; temperature–RH; wind–RH; QNH–wind; RH squared.	Represent nonlinear relationships.	Only include if generated in the actual pipeline.
Sequence features	Ordered windows for LSTM/TFT pathways.	Represent temporal dependencies.	Sequence length, gap policy and split boundaries.
Derived aviation outputs	Headwind/crosswind, RH and density altitude.	Translate predictions into interpretable quantities.	Separate predicted variables from calculated variables.

B.3 Variable units and physical validation

Table B.3: Variable units and physical validation checks

Variable	Unit / convention to state	Suggested data-quality checks
Wind speed	Knots if using the reported wind model metric.	Non-negative values; calm and missing values distinguished.
Wind direction	Degrees clockwise from north where applicable.	Normalize to valid circular domain; special cases documented.
QNH / pressure	hPa if that is the database convention.	Plausibility bounds defined from data/source documentation.
Temperature	°C.	Check missing sentinels and impossible encoded values.
Dewpoint	°C.	Compare with temperature; preserve exceptional cases for review.
Relative humidity	Percent.	Ensure 0–100%; log any clipping or rejection.
Visibility	Source unit and any converted unit.	Special codes, greater-than values and missing cases explicit.
Cloud base	Source unit, commonly feet if decoded that way.	Coverage code and height treated as distinct fields.
Density altitude	Feet if that is the implementation output.	Formula, pressure-altitude input and ISA assumptions recorded.

B.4 Component dataset manifests

Each model run must have a small manifest in order to ensure we do not assume that all of the components were trained on the same number of rows or split protocol.

Table B.4: Component dataset manifest fields

Manifest field	Required entry
Component name	wind; qnh_dewpoint_rh; temperature_pressure_da; cloud_visibility.
Target variable(s)	Explicit variable and unit for each target.
Input columns	Exact ordered feature list or schema file.
Rows before cleaning	Count and source query/version.
Rows after cleaning	Count with exclusion reasons.
Split method	Random or chronological; dates or seed.
Training / validation / test counts	Report separately for each component.
Missingness	Target and input missingness.
Model artefact	Path, version, checksum and training date.
Evaluation artefact	Prediction export and metric script.

B.5 Interpretation of dataset counts

The 121,407 entries and 51,175 three-hour pipeline entries in the database represent distinct processing phases. They cannot be treated as equivalent numbers. The processes of lag generation, rolling window, target shifting, missing value elimination, and sequence formation may decrease the number of data samples available. The thesis needs to mention which number represents the original records and which one represents the processed shared data and the processed training/testing sets for each model.

Appendix C: Additional Model Results and Evaluation Notes

C.1 Reported results from supplied individual material

Results are shown below to the accuracy level possible from the information provided. These are not generated again independently. Metrics for different components cannot be directly compared because they have different targets, units, sample sizes and testing procedures.

Table 0.B.1: Additional model results and interpretation boundaries

Component / model	Reported result	Interpretation boundary / follow-up
Wind XGBoost	MAE 1.0516 knots; RMSE 1.4404 knots; R^2 0.7798; n=10,235 test rows.	Reported 80/20 random split with random_state=42. Add a chronological holdout before claiming future-period generalisation.
Wind training split	40,940 training rows and 10,235 test rows.	These are model-pipeline rows, not the raw database count.
QNH / dewpoint LightGBM	QNH MAE approximately 1.8 hPa; dewpoint MAE approximately 0.58 °C and R^2 approximately 0.66.	Approximate reported values; exact split, test count and metric scripts should be attached.
Derived RH	Calculated from predicted dewpoint and predicted temperature using August–Roche–Magnus relationship.	No separate RH accuracy score was supplied. Evaluate against observed RH on aligned timestamps.
Temperature / pressure RF baseline	R^2 approximately 0.68.	Reported in individual thesis; target aggregation and units require clarification.
Temperature / pressure LSTM baseline	R^2 approximately 0.25.	Reported in individual thesis; sequence setup and evaluation sample details require confirmation.

Temperature / pressure hybrid RF–LSTM	R^2 approximately 0.84; MAE approximately 0.866.	Reported values; clarify whether metrics are per target or aggregated and state units/sample count.
Cloud / visibility Decision Tree	Accuracy 76.25%.	Reported in group paper; target label, class balance and split details should accompany the score.
Cloud / visibility Random Forest	Accuracy 76.31%.	Individual implementation material mentions RandomForestClassifier.
Cloud / visibility XGBoost	Accuracy 78.15%.	Implementation discrepancy remains: paper reports XGBoost, while visible script reportedly uses RandomForestClassifier. Do not attribute score to code until reconciled.

C.2 Metrics to report consistently

Table C.2: Evaluation of metrics and their purposes

Metric	Definition / purpose	Use
MAE	Mean absolute error in target units.	Wind speed, QNH, temperature, dewpoint and continuous visibility.
RMSE	Square root of mean squared error; penalises large errors more.	Continuous prediction targets.
R^2	Explained-variance-style comparison against a mean predictor.	Regression, interpreted alongside MAE/RMSE and evaluation design.

Accuracy	Fraction of correct class predictions.	Cloud/visibility classification, accompanied by class distribution.
Precision	Correct positive predictions divided by predicted positives.	Hazard-relevant cloud or low-visibility class.
Recall	Correct positive predictions divided by actual positives.	Assess whether rare but important classes are missed.
F1-score	Harmonic mean of precision and recall.	Imbalanced classification.
Confusion matrix	Counts of correct and incorrect class assignments.	Classification diagnostics.

C.3 Additional analysis for each model

C.3.1 Wind prediction

Wind component must be able to output separate wind speed and direction. Wind direction is calculated on a circular basis where, for example, 359° and 1° would be 2° apart and not 358°. In case runway-relative outputs are being generated, predicted wind must be converted to headwind and crosswind components using a defined runway heading. XGBoost score is an informative measure; however, it alone does not verify runway alerts nor prove correct detection of all thresholds of operational relevance.

C.3.2 QNH, dewpoint and RH

Because QNH and dewpoint have different units of measurement and different scales of errors, it would be better to use them separately. Relative humidity is a derived value, and its error is a function of both the error in the predicted temperature and the error in the predicted dewpoint. One metric that can be applied for the evaluation is a comparison of the derived relative humidity with the observed one.

C.3.3 Temperature, pressure and density altitude

Comparison between the RF, LSTM and the hybrid methods should only be done when there is a similarity in terms of the target, sample, and metric calculation. Separate values of the MAE and RMSE should be used for temperature and pressure. There are two steps to evaluate density altitude, which include verifying the equation separately and comparing density altitude from forecast input with density altitude from observation input. inputs.

C.3.4 Cloud and visibility

It will have to indicate whether visibility is considered continuous or categorical data, as well as what particular cloud type is being forecasted. Sole reliance on classification accuracy might miss the failure of the algorithm to work properly in low visibility scenarios. Class distribution, confusion matrix, precision, recall, and F1 score should be included wherever possible. This issue regarding code/model in cloud data should be sorted out.

C.4 Result publication checklist

- State the test period and sample count for every metric.
- State units for all continuous targets.
- Use timestamp-aligned prediction/observation pairs.
- Identify the exact model artefact and code version.
- Include a simple baseline where evidence is available.
- Do not invent missing metrics or additional decimal places.
- Separate reported results from newly reproduced results.
- Explain known implementation inconsistencies explicitly.

Appendix D: System Architecture Diagrams

D.1 Logical architecture

Figure provides the logical architecture according to the four parts and process flow illustrated in the provided project material. It is a diagram used for documentation purposes and not a screenshot from a working application.

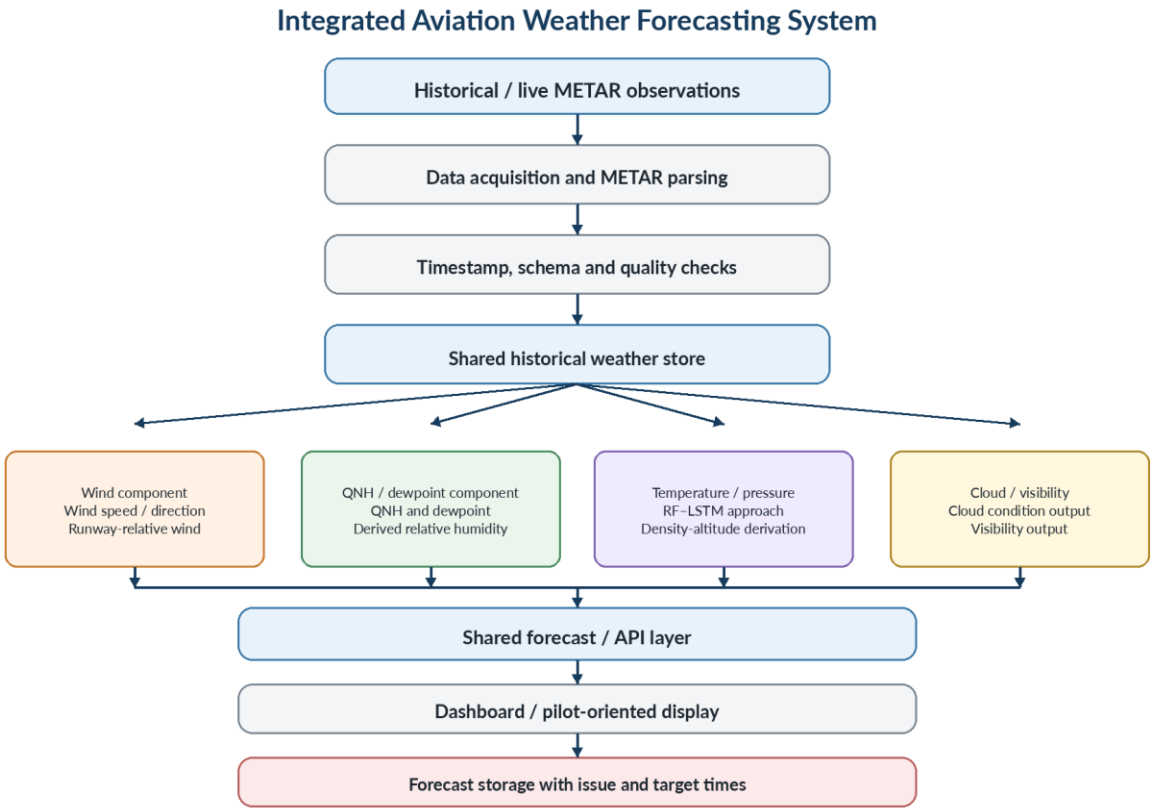


Figure D.1: High-level architecture of the integrated aviation weather forecasting system

D.2 Component interfaces

Table D.1: Component interfaces and integration requirements

Component	Input	Output	Integration requirement
Wind / runway	Wind speed, direction, runway heading and target time.	Wind forecast, headwind/crosswind and any rule-based flag.	Preserve direction convention, units and runway identifier.
QNH / dewpoint / RH	Weather features and aligned temperature prediction for RH.	QNH, dewpoint and calculated RH.	Reject or mark incomplete if required inputs or target times disagree.
Temperature / pressure / DA	Historical features and ordered sequence.	Temperature, pressure and derived density altitude.	Document hybrid combination rule, units and formula version.
Cloud / visibility	Parsed cloud, visibility and relevant weather features.	Explicitly defined class or continuous output.	Document label schema and classifier version.
Verification layer	Stored forecast and later target observation.	Paired records and evaluation metrics.	No score until a valid target-time match exists.

Appendix E: Testing Results and Test Cases

E.1 Test evidence status

However, from the provided documentation, there are descriptions of the components and evaluation criteria but no test log for all the integrated functions. Hence, what follows is a specification of the tests that will be run on the final code; the test log will contain the result, date, tester, and evidence path of each of the following test cases.

E.2 Data acquisition and preprocessing tests

Table E.1: Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
DATA-01	Valid METAR record received.	Record is stored and parsed into the expected fields.	Raw report and parsed output.
DATA-02	Missing or malformed timestamp.	Record is rejected or quarantined with a reason.	Validation log.
DATA-03	Duplicate observation timestamp.	Documented duplicate rule is applied consistently.	Before/after counts.
DATA-04	Missing target variable.	Row is excluded from the relevant target set and exclusion is counted.	Cleaning report.
DATA-05	Irregular time gap before T+3 target.	Timestamp-based matching is used; no incorrect row-shift assumption.	Issue/target pair example.

DATA-06	Rolling feature near dataset start.	Insufficient history is marked missing or handled by a documented rule.	Feature rows and rule.
DATA-07	Feature calculation near train/test boundary.	No future or cross-boundary information leaks into features.	Split-aware test output.
DATA-08	Unit conversion or timestamp normalization.	Values and times remain consistent with documented convention.	Input/output example.

E.3 Wind model and runway interpretation tests

Table E.2: Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
WIND-01	Valid wind speed and direction input.	Model returns a numeric prediction with unit and target time.	API response / prediction row.
WIND-02	Direction near 359° and 1°.	Circular representation or error calculation treats them as adjacent directions.	Unit-test output.
WIND-03	Calm wind or missing direction.	Special case is handled without an invalid directional vector.	Test output and warning.
WIND-04	Runway heading and wind direction supplied.	Headwind/crosswind calculation follows the	Known-case calculation.

		documented sign convention.	
WIND-05	Runway heading missing.	Runway-relative output is withheld or marked incomplete.	Response payload.
WIND-06	Alert threshold boundary.	Boundary comparison follows the configured inclusive/exclusive rule.	Threshold test cases.
WIND-07	No eligible feature row.	Prediction is not fabricated; error/status is explicit.	Service log.
WIND-08	Model artefact unavailable.	Service reports unavailable model and avoids stale silent output.	Fault-injection log.

E.4 QNH, dewpoint and relative humidity tests

Table E.3 : Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
QNH-01	Valid features for T+3 h.	QNH output has hPa unit and correct target time.	Prediction row.
QNH-02	Dewpoint model input contains missing required feature.	Prediction is blocked or explicitly flagged according to policy.	Validation log.
QNH-03	Temperature and dewpoint predictions share target time.	RH is calculated from aligned inputs using the documented formula.	Formula test output.
QNH-04	Temperature and dewpoint target times differ.	RH is withheld or marked invalid; mismatched values are not silently combined.	Integration test.
QNH-05	Calculated RH outside 0–100%.	System logs and applies a documented rejection/clipping policy.	Before/after value and log.
QNH-06	Dewpoint equals temperature.	Formula produces a physically consistent saturation case within numerical tolerance.	Independent calculation.
QNH-07	One required RH input missing.	RH is null/incomplete, not a guessed value.	API payload.
QNH-08	Observed RH available for verification.	Only timestamp-aligned pairs are scored; sample count is shown.	Metric export.

E.5 Temperature, pressure and density-altitude tests

Table E.4: Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
TP-01	Valid sequence window supplied to LSTM path.	Sequence order and expected input shape are preserved.	Model input shape/log.
TP-02	Sequence contains a time gap.	Gap is handled by a documented rule; invalid sequence is not silently treated as regular.	Sequence validation output.
TP-03	RF, LSTM and hybrid predictions produced.	Combination rule matches the implementation documentation.	Per-model and hybrid values.
TP-04	Temperature output returned.	Unit is °C and target time is explicit.	API response.
TP-05	Pressure output returned.	Unit and exact pressure field are explicit.	API response.
TP-06	Known pressure/temperature inputs for density altitude.	Independent calculation matches implementation within a defined tolerance.	Hand-check or reference calculation.

TP-07	Missing pressure or temperature input.	Density altitude is marked unavailable, not calculated from guessed data.	Output and log.
TP-08	Observed target values become available.	Forecast DA is compared with DA derived from observed inputs separately from model metrics.	Paired evaluation export.

E.6 Cloud and visibility tests

Table E.5: Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
CV-01	Standard METAR with visibility and cloud groups.	Parser extracts the documented target fields correctly.	Input and parsed output.
CV-02	Visibility encoded with a special or greater-than value.	Encoding is converted or retained according to documented policy.	Parser unit test.
CV-03	Cloud amount present but cloud base missing.	Coverage and height are not confused; missing base is represented explicitly.	Parsed fields.
CV-04	Malformed cloud group.	Parser records a warning or invalid status instead of inventing a class.	Error log.
CV-05	Unknown present-weather code.	Unknown category is handled by a documented fallback.	Encoder output.
CV-06	Rare low-visibility class.	Evaluation includes class-sensitive recall/precision, not accuracy alone.	Confusion matrix.
CV-07	Model script and reported model label differ.	Release is blocked until model/code/prediction artefacts are reconciled.	Model manifest.
CV-08	Target observation unavailable.	Forecast remains pending and is not included in verification metrics.	Status record.

E.7 End-to-end integration tests

Table E.6: Component and end-to-end test cases

ID	Test scenario	Expected behavior	Evidence to capture
INT-01	One valid source observation flows through parse, feature, model, API and UI.	Identifiers and timestamps remain consistent across stages.	End-to-end trace.
INT-02	Different component outputs have different target times.	UI does not imply they represent the same valid time.	API/UI comparison.
INT-03	Forecast is stored and later observation arrives.	Correct target observation is matched by the documented rule.	Paired database rows.
INT-04	Duplicate verification run.	Metrics do not double-count the same forecast-observation pair.	Metric row IDs.
INT-05	API service unavailable.	User receives a clear error and no fabricated forecast.	Client and server logs.
INT-06	Old model artefact or stale data.	Freshness/version status is visible or request is blocked.	Fault-injection result.
INT-07	Display units compared with API units.	UI labels and converted values match the response contract.	Payload and screenshot.
INT-08	New model version replaces prior model.	Historical forecasts remain linked to the original model version.	Database trace.

Appendix F: Supporting Evidence and Limitations

F.1 Model inventory

Table F.1: supplementary information

Component	Model / method described in supplied material	Output	Evidence limitation to resolve
Wind	XGBoost; a Temporal Fusion Transformer pathway is also described.	Wind speed/direction; runway-relative interpretation.	Clarify which saved artefact serves each production endpoint and report TFT metrics separately.
QNH / dewpoint / RH	Separate LightGBM prediction for QNH and dewpoint; RH derived with August–Roche–Magnus.	QNH, dewpoint and calculated RH.	Add exact feature list, split counts and RH validation result if available.
Temperature / pressure / DA	Random Forest and LSTM hybrid for three-hour prediction.	Temperature, pressure, pressure altitude and density altitude.	Record exact hybrid combination rule, per-target metrics, units and sample counts.
Cloud / visibility	Decision Tree, Random Forest and XGBoost results are reported in the group paper; visible individual implementation references RandomForestClassifier.	Cloud/visibility class or target as actually implemented.	Resolve model mismatch and document exact target labels and class thresholds.

F.2 Evidence limitations in the supplied material

The scores provided for the model in Appendix C are based on the supplied individual reports and not recreated separately in this report. The supplied documentation does not prove the existence of the full integration-test log and application screens. The cloud/visibility model needs to reconcile the reported XGBoost score and the use of Random Forest provided in the individual documentation. These issues are clearly mentioned rather than fabricated codes, images/screenshots, and test results.